Design Fundamentals 4/4

Topics 19–24

19. Rate limiting

- Token bucket, leaky bucket, fixed/sliding window
- Per user/IP/region
- A shared store such as Redis
- DoS protection

20. Logging and monitoring

- Structured logs, metrics (time-series), dashboards, alerts on SLOs, tracing

21. Publish/subscribe pattern

- Topics, publishers, subscribers
- At-least-once delivery means idempotent consumers
- Ordering per partition
- Kafka, Pub/Sub

22. MapReduce

- Map → shuffle → reduce over a distributed file system
- Idempotent tasks
- Data stays where it is and the code moves to it

23. Security and HTTPS

- Symmetric/asymmetric encryption, TLS handshake, certificates and CAs, MITM

24. API design

- Resource-oriented REST or RPC/GraphQL
- Pagination, versioning, idempotency keys, rate limits, clear errors

Recap

- 19. Rate limiting
- 20. Logging and monitoring
- 21. Publish/subscribe pattern
- 22. MapReduce
- 23. Security and HTTPS
- 24. API design